COMPANION TO

Capability Matters: A Casebook

LENS Companion

CONCENTRATION · CROSSWALKS · REFERENCES

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Learning Engineering for Next-Generation Systems · v2.1

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LDT · SCHOOL OF EDUCATION · JOHNS HOPKINS UNIVERSITY

SECTION

How to use this companion

The LENS Companion is the concept-facing reference for the LENS specialisation: what LENS is, what it teaches, and how the casebook's cases line up with the curriculum. It travels with advisory boards, prospective students, recruiting conversations, and program reviews — anyone for whom the question is **what is this concentration?** rather than **where in the casebook is case 137?**.

The companion is in four parts:

I. The Five LENS Competencies — formal names, taglines, main objectives, and the v2.1 subobjective set (1.1 – 5.6). The organising layer over the formal CLOs.

II. The CLOs and the Course Mapping — the five concentration learning objectives, the LDT / LENS course descriptions, the course-to-CLO coverage matrix, and the LEN course list with per-course case counts.

III. Crosswalks — the casebook's induced framework mapped to the canonical five competencies, plus the three-anchor convention every v2 case carries.

IV. The canonical program documents — the literal text of the `lens_program/` source-of-record `.md` shadows, inlined verbatim (status banners and change logs preserved). Included so the companion can travel with advisory boards, prospective students, and external reviewers carrying the authoritative wording in hand.

> **For auditors and reviewers.** The casebook's indexes — every case by primary domain, every case by LENS course — and the full per-case references appendix live in a separate document: **Validation and Audit** (`validation-audit.typ` in the casebook source; `capability-matters-validation-audit.pdf` in the build). That document is built from the same casebook source as the companion, so the two stay in sync; it is the place for the verification and accreditation track.

> **On versioning.** This companion documents the **v2.1** program state (June 2026), in which D3 is renamed **Human-System Collaboration** and moved from position 5; T&E moves to D4; Sociotechnical Constraints moves to D5. The order now reads as the flywheel: see the system → build → integrate humans → measure → deploy. Prior versions are preserved in the program docs' change logs (`lens_program/1_LENS_Five_Competencies.md`).

SECTION

Why a companion — the thesis in brief

Learning engineering is a toolbox, not a single tool. At its centre is a dialogue between two traditions — the engineering disciplines that design, build, and sustain complex systems, and the learning sciences and education that study how people come to know and do. Around that dialogue gather many other fields: cognitive and human-factors engineering, measurement, implementation science, design, data and computation, and the operational domains themselves. The work is plural by design.

Capability is a system parameter.

Every complex system depends on what people can do inside it. That dependency is measurable, designable, and too important to leave to chance.

> **01.** Systems don’t fail because the technology breaks. They fail because someone couldn’t do what the system required.
>
> **02.** We engineer every parameter of a system except the people.
>
> **03.** Capability is not a soft problem. It is a systems engineering problem.
>
> **04.** Decision-grade evidence. Not opinions about training.
>
> **05.** Capability without agency is automation. The goal is operators who can perform, adapt, and lead — not comply.
>
> **06.** Sometimes the constraints make the goal unreachable. Saying so early is a result, not a failure.

Capability lives at the **interface** between what a system requires of its operators and the impact the system has to deliver. When that interface is wrong, the gap may originate in human development, in system design, in organisational performance, or in the interaction among them. Distinguishing which is itself a measurement and engineering problem; the casebook calls it **gap attribution**, and treats it as the diagnostic move that the rest of the discipline depends on.

The premise of the Johns Hopkins School of Education’s Learning Design & Technology program — and of the Learning Engineering for Next-Generation Systems specialisation that has grown out of it — is that the capability parameter is engineerable. Not by accident. Not by declaration. By the same kind of evidence-grounded, methods-grounded, cross-domain discipline that built modern reliability engineering, modern epidemiology, and modern systems safety. The discipline does not yet exist at the scale the evidence demands. LDT and LENS are organised to help build it.

> **The casebook's method, in one sentence.** Each case is an **unpacking**: take a real outcome, recover the capability question behind it, and attribute the gap to human development, system design, or organisational performance. The discipline of unpacking is what LENS teaches; the casebook is its primary instrument. The framework that follows is the analytic lens the unpacking method uses — five competencies, each a way of asking what the work the system requires actually is, and whether the people inside it can do it.

SECTION

The Five LENS Competencies — v2.1

Organising layer over the formal CLOs. Formal names carry program documentation; taglines carry recruiting and the site. Subobjective numbers (1.1, 3.2, …) are stable tags for case studies, course modules, and capstone rubrics. The CLO language in the program documentation remains the language of record.

1 Systems Analysis

SYSTEMS THINKING AND ANALYSIS (CLO-1)

see the whole system

Main objective. Trace how human capability and system performance depend on each other, so interventions target the real problem.

1.1 Decompose system performance requirements into measurable human capability requirements: define what the system demands of its operators and the impact it must deliver.

1.2 Model learning environments and their host operational systems as interacting systems, with defined components, interfaces, and feedback loops.

1.3 Apply systems engineering lifecycle models to scope, sequence, and evaluate capability interventions within and across disparate systems.

1.4 Analyze and communicate human-system interdependencies to identify capability gaps and predict operational impact at scale.

1.5 **Governance-objection diagnostic.** Distinguish a governance objection that good design can dissolve from one that correctly signals the system should not deploy. [v2]

2 Iterative Development

LEARNING ENGINEERING DESIGN AND IMPLEMENTATION (CLO-2)

build, test, refine

Main objective. Design capability interventions through iterative engineering cycles that survive contact with real operational environments.

2.1 Design interventions that integrate learning sciences principles with measurable outcomes and system design constraints.

2.2 Run the iteration cycle: design, instrument, evaluate, refine, and revise the problem framing as evidence accumulates.

2.3 Evaluate evidence-based design strategies for transfer to high-consequence settings at operational speed and scale.

2.4 Construct implementation plans that address adoption, sustainment, and lifecycle integration across organizational and system boundaries.

2.5 **Narrate and defend the design iteration in first person.** Design competence includes not only producing an outcome but rendering the iteration legible. [v2]

3 Human-System Collaboration

HUMAN-SYSTEM COLLABORATION AND ADAPTIVE SYSTEMS (CLO-3)

work together

Main objective. Design human-system partnerships — including but not limited to human-AI — that make people more capable while preserving human agency and the recoverability of the system.

3.1 Design role architectures, interface and alert systems, mode/state transparency, authority gradients, and recoverability mechanisms that engineer collaborative capability at the human-system boundary.

3.2 Evaluate the measured impact of system-mediated work on human performance: the gains the system enables and the risks it introduces — automation bias, cognitive offloading, skill atrophy.

3.3 **Delegation with revocation.** Design the human oversight layer for delegated work; specify in advance the disconfirming evidence that would revoke the delegation. [v2]

3.4 **Collaboration measurement.** Measure the capability of a team or collaboration as a unit of analysis, distinct from any individual operator or any individual system component. [v2]

3.5 Specify learning and capability requirements for systems not yet fielded, working from design artifacts rather than operational experience.

4 Test and Evaluation

DATA, MEASUREMENT, AND EVALUATION (CLO-4)

show what works

Main objective. Produce decision-grade evidence linking learning to operational impact — where decision-grade is a sufficiency judgment under irreducible uncertainty — and tell a learning gap from a system gap.

4.1 Design ethical instrumentation strategies that measure capability at the speed and scale operations require.

4.2 **Gap attribution.** Diagnose the gap: differentiate capability shortfalls rooted in human development from those rooted in system design or organizational performance.

4.3 Construct decision-grade evidence artifacts under irreducible uncertainty: state what is known, what is assumed, and what would change the decision.

4.4 **Judgment under inadequate evidence.** Justify a consequential decision on incomplete and contested evidence; document the basis; name what would change it. [v2]

4.5 Communicate evidence honestly to technical and non-technical stakeholders, including uncertainty and limits of inference.

4.6 **Fairness beyond omission.** Demonstrate that omitting a protected attribute does not establish fairness; analyze competing fairness definitions using demographic-stratified outcome evidence. [v2]

5 Navigating Sociotechnical Constraints

CONTEXT AND DOMAIN FLUENCY (CLO-5)

make it work in the real world

Main objective. Integrate capability interventions into the sociotechnical systems they must live in: the regulatory, organizational, cultural, and technical realities that determine whether good designs survive.

5.1 Analyze the regulatory, organizational, cultural, and technical constraints that shape what can be built, deployed, and sustained in a given context.

5.2 Apply human systems integration frameworks to fit capability interventions to operational environments.

5.3 Leverage prior domain expertise, and respect for others', to elicit and validate specialist knowledge that surfaces constraints early.

5.4 Translate constraints and stakeholder requirements across disciplinary and institutional boundaries into coherent design specifications.

5.5 Anticipate adoption and sustainment barriers, and design interventions that survive them.

5.6 **Cross-regime / platform-dependency governance.** Where capability is deployed on a platform governed by a different regime than the one operating it, design the governance seam as an explicit deliverable. [v2]

See the whole system. Build, test, refine. Work together. Show what works. Make it work in the real world.

SECTION

CLOs and Course Mapping — v2.1

The LDT M.Ed. is a 10-course, 30-credit, online/asynchronous master's degree in the School of Education. LDT comprises three concentrations: Learning Experience Design (LXD, currently operating), AI Leadership in Education (AILE, launching August 2026), and Learning Engineering (LENS, launching August 2026). LENS graduates demonstrate competency across five domains; the CLOs below are the assessable form.

THE FIVE CLOS

CLO-1 · Systems Thinking and Analysis. Decompose system performance requirements into measurable human capability requirements. Apply systems engineering lifecycle models. Analyze interdependencies between human and system performance to identify gaps and predict operational impact at scale.

CLO-2 · Learning Engineering Design and Implementation. Design capability development interventions through iterative engineering cycles. Evaluate evidence-based design strategies for transfer to high-consequence settings. Construct and communicate implementation plans.

CLO-3 · Human-System Collaboration and Adaptive Systems. Design human-system partnerships that make people more capable while preserving human agency. Specify delegation with revocation. Measure collaboration as a unit of analysis.

CLO-4 · Data, Measurement, and Evaluation. Design ethical instrumentation. Construct decision-grade evidence under irreducible uncertainty. Apply gap attribution. Demonstrate fairness beyond omission.

CLO-5 · Context and Domain Fluency. Integrate capability interventions into regulatory, organizational, cultural, and technical realities. Synthesize across boundaries. Anticipate adoption and sustainment barriers, including cross-regime / platform-dependency governance.

COURSE-TO-CLO COVERAGE

Course	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5
LDT F1 (Learning Sciences Studio)		✗			
LDT F2 (Critical Perspectives)					✗
LEN 1 (Principles of LE for Complex Systems)	✗	✗			
LEN 2 (Human-AI Teaming and Adaptive Learning)			✗		
LEN 3 (Learning Engineering Systems)	✗				✗
LEN 4 (Evidence, Analytics, Measurement)				✗	
LEN 6 (Applied Problem Solving in LE)	✗	✗			✗
LEN 10 (Capstone)	✗	✗	✗	✗	✗
LEN 5 (Human Capability Analysis, elec.)	○				○
LEN 7 (Bias, Risk, Governance, elec.)					○

LEN 8 (Knowledge Transfer & Org Learning, elec.)		○			○
LEN 9 (Computational and AI Methods, elec.)			○	○	

✗ = required course; ○ = elective. Foundations + concentration + methods + capstone = 8 required courses (24 credits); 2 electives complete the 30-credit degree.

SECTION

Crosswalks

INDUCED 8 (CASEBOOK SCAFFOLD) → CANONICAL 5 (PROGRAM OF RECORD)

The casebook induces an 8-cluster analytic scaffold from its 205 cases (full account: `casebook/competencies.md`). Those eight cluster cleanly into the canonical five LENS competencies — except for D2 Iterative Development, which is the iteration **method** threaded through the cases rather than its own cluster. The gap is what the editor-commissioned first-person Practice Flywheel accounts will close in the next edition.

Induced cluster (casebook)	Canonical D	Notes
1 Capability requirements under operational reality	D1	Engineered vs. stated requirements; sustainment-as-requirement.
2 Evidence architecture the institution cannot deceive itself with	D4	Measurement gaming; post-deployment surveillance.
3 Interface and role design at the human-automation boundary	D3	Cues, alerts, mode/state transparency, recoverability.
4 Pairing mechanism with authorization	D3	Frontline halt authority; non-punitive reporting; authority gradient.
5 Governance architecture for deployment	D5	Consent, oversight, accountability as design artifacts.
6 Cross-organization and cross-time knowledge transfer	D5	Institution-building after catastrophe; tacit-knowledge survival.
7 Capability under system change and aging assumptions	D5	Legacy assets; multi-layer drift; transitions.
8 Equity and construct definition as design commitments	D4	Construct choice; demographic stratification; fairness.
(none specific)	D2	The iteration **method** — threaded across cases, not its own cluster.

THE CASEBOOK'S THREE ANCHORS PER CASE

Every v2 case carries three anchor codes in its header and metadata:

`induced-anchor` — e.g. `"2.4"` — the casebook's induced 8 / 32 sub-cluster the case evidences (analytic scaffold).

`lens-anchor` — e.g. `"D3/PT5"` — the canonical LENS five domain + problem type (1 – 6) the case is **primary** for.

`clo-anchor` — e.g. `"CLO-3"` — the assessable CLO the case serves as a worked example for (course mapping anchor).

APPENDIX · CLOS AND COURSE COVERAGE

The competencies, the CLOs, and the courses.

COMPETENCY → CONCENTRATION LEARNING OUTCOME

The five capabilities the cases turn on are named, in the curriculum, as five concentration learning outcomes (CLOs):

Systems Analysis	CLO-1 · Systems Thinking and Analysis
Iterative Development	CLO-2 · Learning Engineering Design and Implementation
Test and Evaluation	CLO-3 · Data, Measurement, and Evaluation
Navigating Sociotechnical Constraints	CLO-4 · Context and Domain Fluency
Human-System Collaboration	CLO-3 · Human-System Collaboration and Adaptive Systems

THE LEN COURSES · CASES PER COURSE

Each case names the LEN courses it serves as a worked example for, and most cases serve several. The counts below are therefore not a partition of the 202 cases — they sum to well more than 202, because a single case typically appears under three or four courses.

LEN 1	Principles of learning engineering	36
LEN 2	Human-AI teaming	54
LEN 3	Systems integration	49
LEN 4	Evidence and measurement	67
LEN 5	Capability analysis	86

LEN 6	Applied problem-solving	18
LEN 7	Bias and governance	114
LEN 8	Knowledge transfer	98
LEN 9	Computational methods	30
LEN 10	Capstone studio	19

WHAT THE DISTRIBUTION SHOWS

The distribution is itself diagnostic. The strongly covered topics are the ones the existing case literature most readily supports: bias and governance (LEN 7) and knowledge transfer (LEN 8) lead, with capability analysis (LEN 5) and evidence and measurement (LEN 4) close behind. Across 202 cases the corpus is saturated with examples of what goes wrong when ethics, governance, evidence, and institutional learning are not engineered — and, in the success cases, what it takes to engineer them.

The thinner coverage is equally informative. Computational methods (LEN 9) has the failure modes of algorithmic systems well represented but the computational tools for capability instrumentation under-represented. The capstone (LEN 10) maps to fewer cases than the number of studio projects each cohort undertakes. Systems integration (LEN 3) and applied problem-solving (LEN 6) required focused tagging to reach defensible coverage, surfacing that operational systems-engineering content is under-represented in the published case literature relative to its curricular centrality, and that stakeholder-analysis and problem-framing have not historically been packaged as case material by other disciplines. These are pedagogical gaps the LENS studio sequence is organized to close — not by adding more failure cases, but by having students produce the missing artifacts in real operational settings and entering them into the dataset for the next cohort.

SECTION

The canonical program documents

What follows is the literal text of the LENS program documents of record, included so this companion can travel with advisory boards, prospective students, and external reviewers carrying the authoritative wording in hand. These mirror the `.docx` files maintained in `lens_program/`; any discrepancy between the two should be resolved in favour of the `.docx`, which is the source of record. The status banners and change logs that head and tail each document are reproduced verbatim — they carry the traceability that makes the documents source-of-record.

> **Relationship to Parts I–III.** Parts I, II, and III of this companion paraphrase and re-present portions of Documents 1, 2, and 4 for the companion's compact form. Part IV is the verbatim source; where a reader needs to confirm the exact wording — for a syllabus, a recruiting page, an MHEC filing, a SIG conversation — Part IV is the canonical text to cite.

DOC 1 · THE FIVE LENS COMPETENCIES

STATUS: CURRENT (v2.1, June 2026). Adopted by program owner (J. Diamond) and editor (W. Gray-Roncal). Source of record for case anchors, syllabi, capstone rubrics, and recruiting. Supersedes v2.0 (proposal in `v2_research/01_*`) and v1 (initial). Prior prose preserved in git history; see the Change Log at the end of this document. The companion `.docx` needs a program-owner round-trip from this `.md`.

LENS • Learning Engineering for Next-Generation Systems

DOCUMENT SECTION

The Five LENS Competencies

Formal names, taglines, main objectives, and subobjectives for case studies, recruiting, and curriculum. Organizing layer over the formal CLOs. Formal names carry program documentation; taglines carry recruiting and the site. Subobjective numbers (1.1, 3.2, …) are stable tags for case studies, course modules, and capstone rubrics. The CLO language in the program documentation remains the language of record.

1. Systems Analysis • *see the whole system*

Formal domain: Systems Thinking and Analysis (CLO-1)
Main objective. Trace how human capability and system performance depend on each other, so interventions target the real problem.

- **1.1** Decompose system performance requirements into measurable human capability requirements: define what the system demands of its operators and the impact it must deliver.
- **1.2** Model learning environments and their host operational systems as interacting systems, with defined components, interfaces, and feedback loops.
- **1.3** Apply systems engineering lifecycle models to scope, sequence, and evaluate capability interventions within and across disparate systems.
- **1.4** Analyze and communicate human-system interdependencies to identify capability gaps and predict operational impact at scale.
- **1.5** [v2] **Governance-objection diagnostic.** Distinguish a governance objection that good design can dissolve from one that correctly signals the system should not deploy.

2. Iterative Development • *build, test, refine*

Formal domain: Learning Engineering Design and Implementation (CLO-2)
Main objective. Design capability interventions through iterative engineering cycles that survive contact with real operational environments.

- **2.1** Design interventions that integrate learning sciences principles with measurable outcomes and system design constraints.
- **2.2** Run the iteration cycle: design, instrument, evaluate, refine, and revise the problem framing as evidence accumulates.
- **2.3** Evaluate evidence-based design strategies for transfer to high-consequence settings at operational speed and scale.
- **2.4** Construct implementation plans that address adoption, sustainment, and lifecycle integration across organizational and system boundaries.

2.5 [v2] **Narrate and defend the design iteration in first person.** Design competence includes not only producing an outcome but rendering the iteration legible — the case-unpacking practice is the model.

3. Human-System Collaboration · *work together*

[v2.1 — moved from D5 and broadened from "Machine Teaming and Adaptation." The competency now covers human-system collaboration generally; human-AI teaming is one sub-pattern within it, not a separate domain.]
Formal domain: Human-System Collaboration and Adaptive Systems (CLO-3)
Main objective. Design human-system partnerships — including but not limited to human-AI — that make people more capable while preserving human agency and the recoverability of the system.

3.1 Design role architectures, interface and alert systems, mode/state transparency, authority gradients, and recoverability mechanisms that engineer collaborative capability at the human-system boundary.

3.2 Evaluate the measured impact of system-mediated work on human performance: the gains the system enables and the risks it introduces — automation bias, cognitive offloading, skill atrophy.

3.3 [v2] **Delegation with revocation.** Design the human oversight layer for delegated work; specify in advance the disconfirming evidence that would revoke the delegation.

3.4 [v2] **Collaboration measurement.** Measure the capability of a team or collaboration as a unit of analysis, distinct from any individual operator or any individual system component.

3.5 Specify learning and capability requirements for systems not yet fielded, working from design artifacts rather than operational experience.

4. Test and Evaluation · *show what works*

[v2.1 — was D3 in v1; renumbered to D4. Subobjective 4.2 ("gap attribution") was the existing "diagnose the gap"; the v2 sweep names it explicitly.]
Formal domain: Data, Measurement, and Evaluation (CLO-4)
Main objective. Produce decision-grade evidence linking learning to operational impact — where decision-grade is a sufficiency judgment under irreducible uncertainty — and tell a learning gap from a system gap.

4.1 Design ethical instrumentation strategies that measure capability at the speed and scale operations require.

4.2 **Gap attribution.** Diagnose the gap: differentiate capability shortfalls rooted in human development from those rooted in system design or organizational performance.

4.3 Construct decision-grade evidence artifacts (dashboards, reproducible reports, governance documentation) under irreducible uncertainty: state what is known, what is assumed, and what would change the decision.

4.4 [v2] **Judgment under inadequate evidence.** Justify a consequential decision on incomplete and contested evidence; document the basis; name what would change it.

4.5 Communicate evidence honestly to technical and non-technical stakeholders, including uncertainty and limits of inference.

4.6 [v2] **Fairness beyond omission.** Demonstrate that omitting a protected attribute does not establish fairness; analyze competing fairness definitions using demographic-stratified outcome evidence.

5. Navigating Sociotechnical Constraints · *make it work in the real world*

[v2.1 — was D4 in v1; renumbered to D5.]

Formal domain: Context and Domain Fluency (CLO-5)
Main objective. Integrate capability interventions into the sociotechnical systems they must live in: the regulatory, organizational, cultural, and technical realities that determine whether good designs survive.

5.1 Analyze the regulatory, organizational, cultural, and technical constraints that shape what can be built, deployed, and sustained in a given context.

5.2 Apply human systems integration frameworks to fit capability interventions to operational environments.

5.3 Leverage prior domain expertise, and respect for others', to elicit and validate specialist knowledge that surfaces constraints early.

5.4 Translate constraints and stakeholder requirements across disciplinary and institutional boundaries into coherent design specifications.

5.5 Anticipate adoption and sustainment barriers, and design interventions that survive them.

5.6 [v2] **Cross-regime / platform-dependency governance.** Where capability is deployed on a platform governed by a different regime than the one operating it, design the governance seam as an explicit deliverable.

DOCUMENT SECTION

Crosswalk to Formal CLOs

#	Competency	Tagline	CLO	Formal domain (program docs)
1	Systems Analysis	see the whole system	CLO-1	Systems Thinking and Analysis
2	Iterative Development	build, test, refine	CLO-2	Learning Engineering Design and Implementation
3	Human-System Collaboration	work together	CLO-3	Human-System Collaboration and Adaptive Systems
4	Test and Evaluation	show what works	CLO-4	Data, Measurement, and Evaluation
5	Navigating Sociotechnical Constraints	make it work in the real world	CLO-5	Context and Domain Fluency

Each subobjective unbundles an element of the corresponding CLO. The 7 named v2 additions (1.5, 2.5, 3.3, 3.4, 4.4, 4.6, 5.6) and the explicit renaming of 4.2 (gap attribution) are required elements within the parent CLO. The reframing of "decision-grade evidence" as judgment under irreducible uncertainty is carried at 4.3 and 4.4. If any v2 addition creates friction with documentation of record, drop the number and fold the language back; the scheme renumbers cleanly.

DOCUMENT SECTION

Using the Tags

Each subobjective number is a stable tag. Three immediate uses:
Case studies. Each case in the casebook carries two coordinates: failure mode code(s) from the six-mode taxonomy (T/D/N/H/G/K — what went wrong) and competency tags (what the case teaches). The pairing turns the casebook into a searchable curriculum asset: every course module can pull the cases tagged to its subobjectives.
Capstone rubrics. Rubric lines reference tags directly (the gap-attribution rubric line assesses 4.2), giving the attainment profile its evidence trail.

Recruiting and the site. Taglines and main objectives are the public-facing layer for capabilitymatters.org and program one-pagers; formal names carry course documentation; the CLO table stays in the documentation of record.

DOCUMENT SECTION

The five, in one breath

See the whole system. Build, test, refine. Work together. Show what works. Make it work in the real world.

DOCUMENT SECTION

Change Log

v2.1 — June 2026 (current)

- **D3 reorder + rename.** What was D5 (Machine Teaming and Adaptation) is broadened to **Human-System Collaboration** and moved to position 3. T&E becomes D4 (was D3); Sociotechnical Constraints becomes D5 (was D4). The order now reads as the flywheel: see the system → build → integrate humans → measure → deploy.
- **7 new subobjectives** (v2 sweep additions): 1.5 governance-objection diagnostic; 2.5 narrate/defend design iteration; 3.3 delegation with revocation; 3.4 collaboration measurement; 4.4 judgment under inadequate evidence; 4.6 fairness beyond omission; 5.6 cross-regime / platform-dependency governance.
- **1 renamed subobjective:** 4.2 explicitly named "gap attribution" (was the existing "diagnose the gap").
- **Reframed term:** "decision-grade evidence" taught as a sufficiency judgment under irreducible uncertainty, made explicit at 4.3 and 4.4.
- **New tagline:** D3 carries "work together" (parallel to the other four).
- **What the casebook anchor codes mean.** Under v2.1, a `lens-anchor: "D3/PT5"` resolves to **Human-System Collaboration**, problem-type 5. The casebook field remap from v1 codes (D3↔D4 swap, D5→D3) ships in the corresponding casebook update.

v2.0 — May 2026 (superseded by v2.1; never adopted standalone)

- Drafted in `v2_research/01_LENS_revised_competencies_and_CLOs.md`. Renamed D2 → Learning Engineering Design, D4 → Context & Domain Fluency, D5 → Emerging Systems & Futures. Added the 7 subobjectives above (originally framed as "new CLOs" — refactored to subobjective level in v2.1 since they're sub-CLO-grain). Reframed decision-grade evidence.

v1 — March 2026 (initial)

- Five competencies: Systems Analysis · Iterative Development · Test and Evaluation · Navigating Sociotechnical Constraints · Machine Teaming and Adaptation. Subobjectives 1.1–5.4. Prior prose retrievable via `git log --follow lens_program/1_LENS_Five_Competencies.md`.

DOC 2 · LENS OBJECTIVES AND COURSE MAPPING

STATUS: CURRENT (v2.2, July 2026). v2.2 incorporates the LDT Program Learning Objectives of record (dated 03-06-2025) at full subobjective grain and adds the PLO ↔ CLO support map (Section 4.1). CLO content is unchanged from v2.1 (adopted by program owner J. Diamond and editor W. Gray-Roncal, June 2026). Source of record for syllabi, capstone rubrics, and course-CLO mapping. Companion competency document is `1_LENS_Five_Competencies.md` (v2.1). Prior prose preserved in git history; see the Change Log at the end of this document. The companion `.docx` is regenerated from this `.md`; program-owner round-trip of the v2.1/v2.2 changes remains pending.

Learning Design and Technology — M.Ed. • Johns Hopkins School of Education
LENS: Learning Engineering for Next-Generation Systems
Learning Objectives and Course Mapping
Draft for discussion — Revised July 2026. Maps LDT program and LENS concentration learning objectives to the approved 10-course, 30-credit structure. PLO language follows the LDT Program Learning Objectives of record dated 03-06-2025 (six objective areas with numbered subobjectives), reproduced in full in Section 3. Revision notes at end.

DOCUMENT SECTION

1. Program Context

The LDT M.Ed. is a 10-course, 30-credit, online/asynchronous master's degree in the School of Education. LDT comprises three concentrations: Learning Experience Design (LXD, currently operating), AI Leadership in Education (AILE, launching August 2026), and Learning Engineering (LENS, launching August 2026).
Program structure is fixed by MHEC approval. The total number of courses (10) and the number of concentration-specific courses (3) are not subject to change within the next two to three years. Course content within the concentration, methods, capstone, and elective slots is shapeable.
Central concept: capability. LENS defines capability as the interface between what a system requires of its human operators and the impact of that system. When this interface underperforms, the source may be system design, human development, organizational structure, or their interaction. Engineering this interface, while preserving and developing human agency, is the central problem LENS addresses.
Students as domain experts. LENS students arrive grounded in a professional domain: teaching, training, healthcare, engineering, operations. The program treats this prior expertise as an asset to be leveraged, not a gap to be remediated. Across the curriculum, students use their domain fluency to engage subject-matter experts credibly, and they practice extending that fluency into unfamiliar domains. No individual is expected to hold every competency of the field; LENS develops integrators who can lead and work within learning engineering teams.

DOCUMENT SECTION

2. Program Structure

Block
Courses
Credits
Slots
Foundations (shared LDT)
F1, F2
6
2
Concentration (LENS, sequential)

LEN 1, 2, 3
9
3
Methods (LENS-specific)
LEN 4, 6
6
2
Capstone
LEN 10
3
1
Electives (university-wide, with advisor)
Any 2
6
2
Total
30
10

LENS-designed courses: LEN 1, 2, 3 (concentration), LEN 4, 6 (methods), and LEN 10 (capstone); 7 of 10 courses are LENS-specific. LENS also offers LEN 5, 7, 8, and 9 as electives; LEN 5 (Human Capability Analysis and Requirements) and LEN 9 (Computational and AI Methods) are recommended. Students select two electives from across the university with advisor consultation.

Methods: LEN 6 (Applied Problem Solving in Learning Engineering) and LEN 4 (Evidence, Analytics, and Measurement for High-Consequence Domains). Both are LENS-specific. Previous LDT cohorts completed Introduction to Social Science Research and Program Evaluation; the LENS configuration replaces both with concentration-specific methods courses.

Sequencing: LEN 1 → LEN 2 → LEN 3 must be taken in order. LEN 6 requires the concentration courses or concurrent enrollment. LEN 10 (capstone) is the culminating course.

DOCUMENT SECTION

3. LDT Program Learning Objectives

The following program-level objectives apply across all LDT concentrations (LXD, LENS, AILE). The text below reproduces the LDT Program Learning Objectives of record (03-06-2025) in full, at subobjective grain; subobjective numbers (e.g., PLO-1.3) are stable tags for syllabus and assessment mapping. Upon successful completion of the LDT program, graduates will demonstrate competence in six areas. Every required LENS course addresses at least one PLO.

PLO-1. Theory and Evidence in the Design of Experiences and Solutions. Focusing on the integration of theoretical and empirical evidence from the learning sciences, media, and technology to design effective digital learning experiences and solutions. Learners will be able to:

- 1.1. Describe and differentiate theories and ideas from the learning sciences, motivation, and media/technology.
- 1.2. Use a variety of digital technologies to develop learning experiences or solutions for different learning environments and learners.
- 1.3. Analyze and evaluate design options to identify appropriate and applicable theories and practices to accomplish intended instructional outcomes in a given context and for a learner population.
- 1.4. Justify design and deployment decisions based on theory, ideas, models, or evidence.

- 1.5. Design, develop, and deploy digital education solutions grounded in ideas and theories from the learning sciences, motivation, and media/technology.

PLO-2. Ethical and Human-Centered Learning Solutions. Emphasizing the creation of learning solutions that are ethical, inclusive, and focused on the diverse needs of learners, adhering to best practices and industry standards. Learners will be able to:

- 2.1. Identify and distinguish among diverse learner needs and incorporate these considerations into technology-supported learning experiences and solutions.
- 2.2. Apply systematic models and systems thinking to create ethical, learner-focused digital tools and curricula.
- 2.3. Evaluate learning solutions to ensure they advance equity and inclusivity across varied contexts.
- 2.4. Design and develop accessible, inclusive, and flexible technology-supported learning experiences or solutions.

PLO-3. Collaboration and Leadership in Educational Technology. Focusing on effective collaboration with stakeholders and leadership in the design, implementation, and evaluation of educational technology solutions. Learners will be able to:

- 3.1. Collaborate with various involved and interested parties in a constructive, professional, and respectful way.
- 3.2. Communicate shared visions for technology-supported solutions.
- 3.3. Lead efforts to ensure the quality of technology-supported learning experiences.
- 3.4. Lead ongoing evaluation efforts to continuously improve the selection and implementation of technologies and technology-based solutions.
- 3.5. Collaborate and lead to develop evidence-based solutions to address instructional and/or programmatic needs using a variety of appropriate educational technologies and techniques.

PLO-4. Communication Competence. Communicating design and pedagogical decisions effectively using various media and tailored to diverse audiences. Learners will be able to:

- 4.1. Craft effective messages using appropriate media in written, verbal, and visual modalities.
- 4.2. Articulate and justify pedagogical, design, and development decisions to diverse audiences.
- 4.3. Communicate effectively with all invested parties to facilitate the design and development of learning products and solutions.

PLO-5. Sociocultural Aspects of Educational Technology. Addressing the analysis of sociocultural factors, including power dynamics and privilege, and developing solutions to mitigate inequities in the design and deployment of educational technology and technology-supported solutions. Learners will be able to:

- 5.1. Analyze how socioeconomic systems influence technology adoption and evaluation.
- 5.2. Evaluate power dynamics and privilege affecting educational technology designs and implementations.
- 5.3. Develop solutions to address and prevent injustices through educational technology designs and implementations.
- 5.4. Design and develop equitable, inclusive, and accessible technology-supported systems, processes, and products.

PLO-6. Data-Informed Learning Design, Implementation, and Evaluation. Using data to inform the design, development, and evaluation of learning experiences and technology-supported solutions, ensuring continuous improvement through evidence-based decision-making. Learners will be able to:

- 6.1. Collect data using a variety of evaluation and research methods to gain insights into learner and educator needs and tool effectiveness.
- 6.2. Use data to revise designs, practices, and goals based on evaluation data for continuous improvement.
- 6.3. Analyze educational data using qualitative, quantitative, and mixed methods.
- 6.4. Design and implement comprehensive data collection and analysis efforts to inform decisions concerning design, development, delivery, and evaluation of programs, processes, and/or products.

DOCUMENT SECTION

4. LENS Concentration Learning Objectives

LENS graduates demonstrate competency across five domains. Each concentration-level objective (CLO) is written at the Analyze, Evaluate, or Create level of Bloom's taxonomy and is structured for direct assessment. Four threads are woven across all CLOs: evidence and measurability; capability (the interface between system requirements and system impact); iterative engineering cycles at speed and scale; and communication across disciplinary and institutional boundaries. A fifth commitment, the preservation and development of human agency, serves as a design constraint across every CLO. The adoption, sustainment, and lifecycle-integration language in CLO-2 and CLO-5 is the concentration's implementation-science commitment made assessable: interventions are engineered to survive contact with the organizations that must carry them, and the implementation evidence feeds the next iteration (Section 6.4 traces the thread course by course).

The five domains align with the learning engineering competency framework under development with the IEEE ICICLE community. Elements not yet field consensus (gap attribution in CLO-4; sociotechnical integration in CLO-5; human-system collaboration including human-AI teaming in CLO-3; the v2 additions on delegation with revocation, judgment under inadequate evidence, fairness beyond omission, and cross-regime governance) are LENS's proposed extensions to that framework, and LENS capstone evidence is offered toward their validation.

Each CLO carries an accessible competency name and tagline used in recruiting and the casebook: CLO-1 Systems Analysis (see the whole system); CLO-2 Iterative Development (build, test, refine); CLO-3 Human-System Collaboration (work together); CLO-4 Test and Evaluation (show what works); CLO-5 Navigating Sociotechnical Constraints (make it work in the real world). Each CLO decomposes into numbered subobjectives (e.g., 4.2 gap attribution, 3.3 delegation with revocation) that serve as stable tags for case studies, course modules, and capstone rubrics; the subobjective set is maintained in the companion competencies document.

ID

Competency Domain

Graduates will be able to...

CLO-1

Systems Thinking and Analysis

Decompose system performance requirements into measurable human capability requirements, defining the interface between what a system requires of its operators and the impact it must deliver. Model learning environments and their host operational systems as interacting systems with defined components, interfaces, and feedback loops. Apply systems engineering lifecycle models to scope, sequence, and evaluate capability development interventions within and across disparate systems. Analyze and communicate interdependencies between human performance and system performance to identify capability gaps and predict operational impact at scale. [v2] Apply the governance-objection diagnostic to distinguish a governance objection that good design can dissolve from one that correctly signals the system should not deploy.

CLO-2

Learning Engineering Design and Implementation
Design capability development interventions through iterative engineering cycles (design, instrument, evaluate, refine) that integrate learning sciences principles with measurable outcomes and system design constraints, and that function at speed and scale within operational environments. Evaluate the operational impact of evidence-based design strategies for transfer to high-consequence settings. Construct and communicate implementation plans that address adoption, sustainment, and lifecycle integration across organizational and system boundaries. [v2] Narrate and defend the design iteration in first person; design competence includes rendering iteration legible, not only producing an outcome.
CLO-3
Human-System Collaboration and Adaptive Systems
Design human-system partnerships — including but not limited to human-AI — that make people more capable while preserving human agency and the recoverability of the system. Design role architectures, interface and alert systems, mode/state transparency, authority gradients, and recoverability mechanisms that engineer collaborative capability at the human-system boundary. Evaluate the measurable impact of system-mediated work on human performance, assessing both the gains the system enables and the risks it introduces, including automation bias, cognitive offloading, and skill atrophy. [v2] Specify delegation with revocation: design the human oversight layer for delegated work and specify in advance the disconfirming evidence that would revoke the delegation. [v2] Measure the capability of a team or collaboration as a unit of analysis, distinct from any individual operator or any individual system component. Communicate evidence-based recommendations for when system augmentation improves versus degrades capability outcomes and operator agency. Specify capability requirements for systems not yet fielded, working from design artifacts.
CLO-4
Data, Measurement, and Evaluation
Design ethical instrumentation strategies that produce measurable evidence of capability (the interface between operator requirements and system impact) at the speed and scale operations require. Construct decision-grade evidence artifacts (dashboards, reports, governance documentation) that link learning outcomes to operational impact and feed subsequent design iterations — where decision-grade is a sufficiency judgment under irreducible uncertainty, naming what is known, what is assumed, and what would change the decision. [v2] Justify a consequential decision on incomplete and contested evidence and document the basis. **Gap attribution.** Differentiate capability gaps attributable to human development from those attributable to system design or organizational performance, and communicate findings to technical and non-technical stakeholders. [v2] **Fairness beyond omission.** Demonstrate that omitting a protected attribute does not establish fairness; analyze competing fairness definitions using demographic-stratified outcome evidence.
CLO-5
Context and Domain Fluency
Integrate capability interventions into the sociotechnical systems they must live in: the regulatory, organizational, cultural, and technical realities that determine whether good designs survive. Analyze the constraints that shape what can be built, deployed, and sustained in healthcare, defense, or education contexts. Apply human systems integration frameworks to fit capability development approaches to operational environments. Leverage prior domain expertise, and respect for others', to elicit and validate specialist knowledge that surfaces constraints early. Synthesize and communicate stakeholder requirements across disciplinary and institutional boundaries into coherent design specifications, and design interventions that survive adoption and sustainment barriers. [v2] **Cross-regime / platform-dependency governance.** Where capability is deployed on a platform governed by a different regime than the one operating it, design the governance seam as an explicit deliverable.

4.1 PLO ↔ CLO Support Map

How the LENS concentration objectives support the six LDT program objectives (03-06-2025). ● = primary support (the CLO's core work directly develops the PLO area); ○ = supporting (the CLO develops the PLO area in the LENS inflection — high-consequence domains, capability engineering, decision-grade evidence).

	PLO-1 Theory & Evidence	PLO-2 Ethical & Human-Centered	PLO-3 Collaboration & Leadership	PLO-4 Communication	PLO-5 Sociocultural Aspects	PLO-6 Data-Informed
CLO-1 Systems Analysis	●	○	○	○		●
CLO-2 Iterative Development	●	●	○	○		●
CLO-3 Human-System Collaboration	○	●	●	○	○	○
CLO-4 Test and Evaluation	○	●	○	●	●	●
CLO-5 Sociotechnical Constraints	○	●	●	●	●	○

Reading the map at subobjective grain:

- **PLO-1 (Theory and Evidence)** is carried primarily by CLO-1 (decomposing system requirements into measurable human capability requirements grounds design in evidence; PLO-1.3, 1.4) and CLO-2 (iterative engineering cycles integrating learning sciences principles; PLO-1.1, 1.2, 1.5), with CLO-4 supplying the evidence side of "justify design and deployment decisions" (PLO-1.4).

- **PLO-2 (Ethical and Human-Centered)** is supported across all five CLOs; primary weight sits in CLO-2 (systematic models and systems thinking, PLO-2.2), CLO-3 (preserving human agency as a design constraint; diverse operator needs, PLO-2.1, 2.4), and CLO-4 (ethical instrumentation; evaluating solutions for equity, PLO-2.3, reinforced by subobjective 4.6 fairness beyond omission).

- **PLO-3 (Collaboration and Leadership)** is carried primarily by CLO-3 (collaboration engineered as a measurable property of human-system teams; PLO-3.1, 3.5) and CLO-5 (synthesizing stakeholder requirements across disciplinary and institutional boundaries; leading adoption and sustainment, PLO-3.2, 3.3, 3.4). PLO-3.4's continuous-improvement leadership is where the implementation-science thread (Section 6.4) meets the program objectives directly.

- **PLO-4 (Communication Competence)** is threaded through every CLO's communication verbs; primary weight sits in CLO-4 (decision-grade evidence artifacts for technical and non-technical audiences; PLO-4.1, 4.2) and CLO-5 (communicating design specifications across boundaries; PLO-4.3).

- **PLO-5 (Sociocultural Aspects)** is carried primarily by CLO-5 (regulatory, organizational, and cultural constraints; power dynamics of deployment contexts; PLO-5.1, 5.2) and CLO-4 (demographic-stratified outcome evidence and fairness analysis; PLO-5.3, 5.4 via subobjective 4.6), with CLO-3's agency-preservation commitment supporting PLO-5.3.

- **PLO-6 (Data-Informed Design, Implementation, and Evaluation)** is carried primarily by CLO-4 (instrumentation, measurement, and evaluation at operational speed and scale; PLO-6.1–6.4), CLO-2 (the evaluate-refine half of the iteration cycle; PLO-6.2), and CLO-1 (measurable capability requirements as the object of data collection; PLO-6.1). PLO-6.2 — using data to revise designs and practices for continuous improvement — is the implementation-science thread's second direct anchor in the program objectives.

Every PLO area receives primary support from at least one CLO, and every PLO subobjective family is addressed by at least one required LENS course (Section 6.1). The LENS inflection is consistent throughout: the same six program areas, developed in high-consequence operational contexts where evidence must be decision-grade and capability is the interface between operator requirements and system impact.

DOCUMENT SECTION

5. Course Descriptions and Objectives Mapping

5.1 Foundation Courses (shared LDT, 6 credits)

LDT F1: Learning Sciences Studio
Foundation (Required, Shared) • 3 credits
Introduces students to major theories of learning (behavioral, cognitive, and situative) and motivation, emphasizing practical application to educational technology and technology-supported learning environments. Learners engage in theory-to-practice activities, critique existing technologies, and collaboratively develop design proposals for new learning environments. Blends theoretical insight with real-world problem-solving to develop practical skills in aligning instructional strategies and tools with diverse learner needs and settings. Prerequisite: none.
Program Objectives (PLO)
Concentration Objectives (CLO)
PLO-1, PLO-2
CLO-2
LDT F2: Critical Perspectives on Educational Technology
Foundation (Required, Shared) • 3 credits
Students analyze educational technologies through the lenses of critical theory and critical theory of technology. Examines the intersections of power, privilege, and societal structures in the design and deployment of learning technologies. Students deconstruct prevailing narratives, investigate how digital tools mediate educational experiences and shape social norms, equity, and access, and develop skills to dissect and challenge assumptions, fostering a nuanced understanding of the ethical, cultural, and political dimensions of technology-infused education. Prerequisite: Learning Sciences Studio.
Program Objectives (PLO)
Concentration Objectives (CLO)
PLO-2, PLO-5
CLO-5

5.2 Concentration Courses (LENS sequential core, 9 credits)

LEN 1: Principles of Learning Engineering for Complex Systems
Concentration 1 (Required, Sequential) • 3 credits
First course in the LENS concentration sequence. Establishes the intellectual and methodological principles of learning engineering as a systems-level discipline. Introduces capability as the interface between what a system requires of its human operators and the impact of that system, and frames the engineering of this interface as the central problem of the field. Examines the convergence of learning sciences, human systems integration, and systems engineering when human capability is treated as a design problem within and across disparate systems. Core topics: defining measurable capability requirements; systems lifecycle thinking; system integration and interdependencies; the iterative engineering cycle (design, instrument, evaluate, refine) as the field's core method; the evidence-capability-performance flywheel (faster evidence cycles accelerate capability development at speed and scale); ethics of intervention where human and system performance are interdependent. Introduces the professional dispositions of the field: focus on learning and learners, reflective and iterative practice, responsiveness to evidence, systems thinking, and mindfulness of context. Students practice communicating capability analyses to diverse stakeholders.
Program Objectives (PLO)
Concentration Objectives (CLO)
PLO-1, PLO-2, PLO-3
CLO-1, CLO-2
LEN 2: Human-AI Teaming and Adaptive Learning Systems
Concentration 2 (Required, Sequential) • 3 credits
Second course in the LENS concentration sequence. Examines human-AI teaming as a design challenge for capability development. Students explore AI as a creative partner (generating, adapting, and evaluating

learning interventions) while assessing the measurable impacts of AI on human performance. Design work proceeds through rapid iteration: students prototype, instrument, and refine AI-mediated learning configurations against measured human performance. Develops evidence-based frameworks for reasoning about AI-mediated learning under uncertainty, with explicit attention to automation bias, cognitive offloading, and the conditions under which AI augmentation helps versus harms. Students communicate AI risk and opportunity assessments to non-technical stakeholders. Builds on the systems principles established in LEN 1.

Program Objectives (PLO)
Concentration Objectives (CLO)
PLO-1, PLO-2, PLO-4, PLO-5
CLO-3

LEN 3: Learning Engineering Systems
Concentration 3 (Required, Sequential) • 3 credits

Third course in the LENS concentration sequence. Applies human systems integration frameworks, systems engineering principles, and sociotechnical analysis to capability development in high-consequence operational environments. Healthcare, defense, and education are treated as co-equal domains. Students model capability development challenges as interacting systems, mapping components, interfaces, and feedback loops before intervening, and address system integration challenges where capability development must function within and across disparate systems at the speed and scale that operational contexts demand. Students analyze the measurable impact of capability interventions on operational outcomes, navigate regulatory, organizational, and cultural contexts, and communicate system-level design decisions and evidence across disciplinary and institutional boundaries. Integrates systems thinking from LEN 1 with the AI teaming frameworks from LEN 2.

Program Objectives (PLO)
Concentration Objectives (CLO)
PLO-2, PLO-3, PLO-4, PLO-5
CLO-1, CLO-5

5.3 Methods Courses (LENS-specific, 6 credits)

LEN 6: Applied Problem Solving in Learning Engineering
Methods 1 (Required) • 3 credits

LENS-specific methods course. Students work authentic cross-domain problems in learning engineering through structured analytical methods. Develops problem framing, stakeholder analysis, needs assessment, and initial evidence-gathering approaches to capability challenges across healthcare, defense, and education contexts. Students apply systems framing to decompose each problem and carry it through at least one full iteration cycle, revising the problem framing as evidence accumulates. Throughout, students leverage their existing professional domain expertise to engage subject-matter experts: eliciting specialist knowledge, validating problem framings with experts, and translating expert knowledge into design requirements. Students communicate problem analyses and preliminary impact assessments to diverse audiences. Introduces the methodological vocabulary and structured reasoning that deeper analytical methods in LEN 5 (recommended elective) build upon. Prerequisite: concentration courses (LEN 1–3) or concurrent enrollment.

Program Objectives (PLO)
Concentration Objectives (CLO)
PLO-1, PLO-3, PLO-4, PLO-6
CLO-1, CLO-2, CLO-5

LEN 4: Evidence, Analytics, and Measurement for High-Consequence Domains
Methods 2 (Required) • 3 credits

LENS-specific methods course developing competence in ethical measurement, learning analytics, and evidence synthesis for high-consequence operational environments. Students operationalize capability constructs, design data collection strategies that function at speed and scale across disparate systems, and combine qualitative and quantitative methods to produce decision-grade evidence linking capability (the operator-system interface) to operational impact. Core topics: ethical instrumentation and data governance; gap attribution (distinguishing capability gaps rooted in human development from those

rooted in system design or organizational performance); learning analytics pipelines that sustain rapid evidence cycles and close the loop between measurement and design revision; communicating evidence to technical and non-technical decision-makers.
Program Objectives (PLO)
Concentration Objectives (CLO)
PLO-2, PLO-4, PLO-6
CLO-4

5.4 Capstone (3 credits)

LEN 10: Learning Engineering Project
Capstone (Required) • 3 credits
End-to-end learning engineering project in a real organizational context. Students identify a capability development challenge at a partner organization (APL as founding partner pathway), gather measurable evidence, design and evaluate an intervention, and assess operational impact. Projects must document at least one complete iteration cycle (design, instrument, evaluate, refine) in the reproducible report. Students are encouraged to site their project in a domain where they hold prior expertise, using that fluency to engage organizational experts credibly while practicing the boundary-crossing communication the field requires. Projects address system integration, speed, and scalability considerations appropriate to the partner context. Integrates all five competency domains. Signature deliverables communicate findings to multiple audiences: evidence dashboard, reproducible report, and governance/ethics plan. Portfolio-quality standards required.
Program Objectives (PLO)
Concentration Objectives (CLO)
PLO-2, PLO-3, PLO-4, PLO-6
CLO-1, CLO-2, CLO-4, CLO-5, CLO-3

5.5 LENS Electives (students choose 2; LEN 5 and LEN 9 recommended)

LEN 5: Human Capability Analysis and Requirements
Elective (Recommended) • 3 credits
Recommended elective providing the deep analytical methods central to LENS practice. Covers cognitive task analysis, critical decision method, capability requirements elicitation, and human factors methods applied to learning system design. Students practice expert knowledge elicitation directly, leveraging their own domain backgrounds to establish credibility with subject-matter experts and extending elicitation skills into unfamiliar domains. Students conduct structured analyses in operational settings, translate findings into measurable design requirements, assess the operational impact of capability gaps, and communicate results to interdisciplinary teams. Builds on the problem-framing methods established in LEN 6.
Program Objectives (PLO)
Concentration Objectives (CLO)
PLO-1, PLO-6
CLO-1, CLO-5
LEN 7: Bias, Risk, and Governance in Learning System Design
Elective • 3 credits
Deepens the equity and accountability thread that runs across the curriculum. Examines algorithmic bias, surveillance, institutional accountability, and governance structures for capability development systems in regulated and high-consequence environments. Students design governance plans, conduct measurable risk assessments for learning technology deployments, and evaluate the equity impact of capability development interventions on affected populations.
Program Objectives (PLO)
Concentration Objectives (CLO)
PLO-2, PLO-5
CLO-5
LEN 8: Knowledge Transfer and Organizational Learning
Elective • 3 credits
Deepens the implementation science thread that runs across the curriculum. Covers knowledge transfer mechanisms, organizational learning theory, adoption and sustainment of capability development

interventions, and implementation science frameworks (EPIS, CFIR, Active Implementation). Students analyze real organizational knowledge transfer problems, design evidence-based sustainment strategies, evaluate the operational impact of implementation approaches, and communicate recommendations to organizational leadership.
Program Objectives (PLO)
Concentration Objectives (CLO)
PLO-3, PLO-4
CLO-2, CLO-5
LEN 9: Computational and AI Methods for Learning Engineers
Elective (Recommended) • 3 credits
Recommended elective providing advanced computational methods applied to learning engineering: natural language processing, machine learning for learner modeling, LLM-as-judge evaluation designs, and computational analytics for capability development data. Students build and evaluate computational tools that serve as creative partners in learning system design, instrumentation, and analysis. Assess the measurable impact of computational approaches on evidence quality and communicate technical findings to non-technical audiences. Complements the measurement methods of LEN 4 with computational depth.
Program Objectives (PLO)
Concentration Objectives (CLO)
PLO-1, PLO-4, PLO-6
CLO-4, CLO-3

DOCUMENT SECTION

6. Coverage Matrix

Verifies that every PLO and CLO is addressed by required courses (foundations + concentration + methods + capstone = 8 courses). ✘ = required; ○ = elective.

6.1 Program Learning Objectives

Course
PLO-1
PLO-2
PLO-3
PLO-4
PLO-5
PLO-6
LDT F1
✘
✘
LDT F2
✘
✘
LEN 1
✘
✘
✘
LEN 2
✘
✘
✘
✘
LEN 3
✘
✘
✘
✘
LEN 6

✘
✘
✘
✘
LEN 4
✘
✘
✘
LEN 10
✘
✘
✘
✘
LEN 5
○
○
LEN 7
○
○
LEN 8
○
○
LEN 9
○
○
○

6.2 Concentration Learning Objectives

Coverage of each CLO by course. ✘ = required course; ○ = elective. Column order matches v2.1 (D3 Human-System Collaboration sits at position 3).

Course	CLO-1 Systems Analysis	CLO-2 Iterative Dev	CLO-3 Human-System Collaboration	CLO-4 Test & Evaluation	CLO-5 Sociotechnical Constraints
LDT F1		✘			
LDT F2					✘
LEN 1	✘	✘			
LEN 2			✘		
LEN 3	✘				✘
LEN 6	✘	✘			✘
LEN 4				✘	
LEN 10	✘	✘	✘	✘	✘
LEN 5 (elec.)	○				○
LEN 7 (elec.)					○
LEN 8 (elec.)		○			○
LEN 9 (elec.)			○	○	

6.3 Coverage Analysis

PLO coverage: All six program objectives are addressed by multiple required courses. PLO-1 (Theory and Evidence): F1, LEN 1, LEN 2, LEN 6. PLO-2 (Ethical and Human-Centered): F1, F2, LEN 1, LEN 2, LEN 3, LEN 4, LEN 10. PLO-3 (Collaboration and Leadership): LEN 1, LEN 3, LEN 6, LEN 10. PLO-4 (Communication Competence): LEN 2, LEN 3, LEN 4, LEN 6, LEN 10; threaded across the concentration, methods, and capstone. PLO-5 (Sociocultural Aspects): F2, LEN 2, LEN 3. PLO-6 (Data-Informed): LEN 4, LEN 6, LEN 10.

CLO coverage: All five competency domains are addressed by required courses. CLO-1 (Systems Thinking and Analysis) is now carried by four required courses (LEN 1, LEN 3, LEN 6, LEN 10), reflecting its weight in the concentration. CLO-4 (Data, Measurement, and Evaluation) is strongly covered through LEN 4 (required methods) and LEN 10 (capstone). CLO-3 (Human-System Collaboration) is owned by LEN 2, with additional depth through LEN 9 (recommended elective). The capstone (LEN 10) integrates all five domains. Iterative engineering cycles appear as required content in LEN 1 (introduced as core method), LEN 2 (prototype-instrument-refine), LEN 6 (full iteration on an authentic problem), LEN 4 (evidence pipelines closing the measurement-to-revision loop), and LEN 10 (documented complete cycle).

6.4 Cross-Cutting Threads

Four commitments are threaded across all courses rather than housed in dedicated courses:

Equity and accountability as design commitments: Grounded in F2 (Critical Perspectives on Educational Technology), threaded through all concentration and methods courses. LEN 4 adds ethical instrumentation and data governance. LEN 7 (elective) provides deeper treatment.

Implementation science (adoption, sustainment, full lifecycle): the through-line that carries capability interventions from evidence to sustained practice. Grounded in LEN 1 (the evidence-capability-performance flywheel), applied in LEN 3 (adoption, sustainment, and lifecycle integration in operational environments) and LEN 6 (problem framings revised as implementation evidence accumulates), and required in LEN 10, where the capstone's reproducible report and governance/ethics plan must address adoption and sustainment in the partner context. LEN 8 (elective) provides the deep treatment: implementation science frameworks (EPIS, CFIR, Active Implementation) applied to real organizational knowledge-transfer problems. The thread directly supports PLO-3.4 (leading ongoing evaluation to continuously improve the selection and implementation of technologies) and PLO-6.2 (using data to revise designs, practices, and goals for continuous improvement), and it is the curricular answer to the casebook's central implementation finding: the 17-year research-practice gap and the training-transfer literature both document that evidence does not move itself.

AI as both creative partner and epistemic risk: Addressed explicitly in LEN 2 (required). LEN 9 (recommended elective) provides computational depth. Attention throughout the sequence to guarding against offloading thinking.

Professional dispositions of the field: The five dispositions named in the field's foundational competencies work (focus on learning and learners; reflective practitioner and iterative refiner; responsive to data, evidence, research, and requirements; leverage systems thinking; mindful of context) are introduced in LEN 1 and reinforced across the sequence. The iterative-refiner and systems-thinking dispositions appear as required content in every LENS course.

6.5 Elective Advising Guidance

LENS students select two electives from anywhere in the university with advisor consultation. LENS-designed electives (LEN 5, 7, 8, 9) are available in the SOE catalogue. LEN 5 (Human Capability Analysis and Requirements) and LEN 9 (Computational and AI Methods for Learning Engineers) are recommended.

Suggested pairings by background:

Engineers and technical professionals entering learning design: LEN 5 (Human Capability Analysis) + LEN 7 (Bias, Risk, and Governance). Builds analytical depth while strengthening contextual and organizational fluency.

Education and training professionals moving toward systems practice: LEN 5 (Human Capability Analysis) + LEN 9 (Computational and AI Methods). Deepens both the analytical and computational competencies that distinguish LENS graduates.

Students pursuing computational depth: LEN 9 (Computational and AI Methods) + LEN 5 (Human Capability Analysis). A Whiting School data science elective may substitute for one LENS elective with advisor approval.

Students with strong methods backgrounds: LEN 8 (Knowledge Transfer and Organizational Learning) + LEN 7 (Bias, Risk, and Governance). Strengthens implementation and governance competencies for students who already bring analytical methods from prior training.

These are recommendations, not requirements. Students may select any university electives that serve their professional goals; LENS advisors work with each student to identify the best fit.

7. Revision Notes (June 2026)

This revision responds to reviewer feedback on three points, plus alignment with the field competency framework in development with the IEEE ICICLE community.

Iterative design strengthened. The iteration cycle (design, instrument, evaluate, refine) is now named in CLO-2, introduced as the field's core method in LEN 1, practiced in LEN 2 and LEN 6, sustained through LEN 4's evidence pipelines, and required as a documented complete cycle in the LEN 10 capstone. Previously iteration appeared only implicitly via the LEN 1 flywheel.

Systems thinking extended into methods. CLO-1 adds explicit system modeling language (components, interfaces, feedback loops). LEN 3 and LEN 6 now require systems framing before intervention; LEN 6 maps to CLO-1, raising required-course coverage of CLO-1 from three courses to four.

CLO-5 reframed to sociotechnical integration. CLO-5 is renamed Navigating Sociotechnical Constraints. Its objective now leads with integrating interventions into the regulatory, organizational, cultural, and technical systems they must survive; prior domain expertise is retained as the means of surfacing constraints early rather than as the headline. Supporting language remains in Program Context, LEN 6, LEN 5, and the LEN 10 capstone.

Field alignment noted. Section 4 states the CLOs' relationship to the emerging field competency framework and labels gap attribution, domain fluency, and human-AI teaming as LENS's proposed extensions. Section 6.4 adds the field's five professional dispositions as a fourth cross-cutting commitment.

DOCUMENT SECTION

Change Log

v2.2 — July 2026 (current)

- **PLO source refresh.** Section 3 now reproduces the LDT Program Learning Objectives of record (dated 03-06-2025) in full, at subobjective grain (PLO-1.1 through PLO-6.4), replacing the summarized "Includes:" digests. PLO-5 and PLO-6 titles corrected to the full forms ("Sociocultural Aspects of Educational Technology"; "Data-Informed Learning Design, Implementation, and Evaluation"). The prior "verify against the MHEC filing" caveat is retired — the 03-06-2025 list is the source.

- **PLO ↔ CLO support map added** (Section 4.1): primary/supporting support of each program objective by each concentration objective, read at subobjective grain, with the coverage claim (every PLO area has primary support from at least one CLO; every subobjective family is addressed by at least one required course).

- **No CLO content changes.** The five CLOs, their subobjectives, and all course mappings are unchanged from v2.1.

- **Implementation science alignment strengthened.** Section 4 names the adoption/sustainment/lifecycle language in CLO-2 and CLO-5 as the concentration's implementation-science commitment made assessable. Section 6.4 traces the thread course by course (LEN 1 flywheel → LEN 3 and LEN 6 application → LEN 10 capstone requirement → LEN 8 deep treatment, with EPIS/CFIR/Active Implementation) and anchors it to PLO-3.4 and PLO-6.2, the two program subobjectives that name implementation and continuous improvement directly. No PLO/CLO mappings change; the §4.1 support map gains the two anchor notes.

v2.1 — June 2026 (superseded by v2.2; CLO content still current)

- **D3/D4/D5 reorder + CLO-3 rename + renumber.** What was D5/CLO-5 (Machine Teaming and Adaptation / Human-AI Teaming and Adaptive Systems) is broadened to **D3/CLO-3 Human-System**

Collaboration and Adaptive Systems and moved to position 3. T&E becomes D4/CLO-4 (was D3/CLO-3); Sociotechnical Constraints becomes D5/CLO-5 (was D4/CLO-4). The new order reads as the flywheel: see the system → build → integrate humans → measure → deploy.

- **CLO mapping updated for every course** (Sections 5 and 6.2). Map: old CLO-3 → new CLO-4, old CLO-4 → new CLO-5, old CLO-5 → new CLO-3.
- LEN 2 maps to **CLO-3** (was CLO-5). The course title and prose still emphasize human-AI teaming; the v2.1 framing places it within the broader Human-System Collaboration competency. A title broadening to "Human-System Collaboration and Adaptive Learning Systems" would match the v2.1 scope but is deferred to the program owner's call.
- LEN 4 maps to **CLO-4** (was CLO-3). LEN 7 to CLO-5 (was CLO-4). LEN 8 to CLO-2, CLO-5 (was CLO-2, CLO-4). LEN 9 to CLO-4, CLO-3 (was CLO-3, CLO-5). LDT F2 to CLO-5 (was CLO-4). LEN 3 to CLO-1, CLO-5 (was CLO-1, CLO-4). LEN 6 to CLO-1, CLO-2, CLO-5 (was CLO-1, CLO-2, CLO-4). LEN 5 to CLO-1, CLO-5 (was CLO-1, CLO-4).
- **7 named subobjectives** (v2 additions; details in `1_LENS_Five_Competencies.md` v2.1): 1.5 governance-objection diagnostic; 2.5 narrate/defend design iteration; 3.3 delegation with revocation; 3.4 collaboration measurement; 4.4 judgment under inadequate evidence; 4.6 fairness beyond omission; 5.6 cross-regime / platform-dependency governance.
- **Reframed term:** "decision-grade evidence" taught as a sufficiency judgment under irreducible uncertainty.
- **What the casebook anchor codes mean.** Under v2.1, a `lens-anchor: "D3/PT5"` resolves to **Human-System Collaboration**, problem-type 5; a `clo-anchor: "CLO-4"` resolves to **Test and Evaluation**. The casebook field remap from v1 codes ships in the corresponding casebook update.

v2.0 — May 2026 (superseded; never adopted standalone)

- Drafted in `v2_research/01_LENS_revised_competencies_and_CLOs.md`. The 7 subobjectives above were originally framed as "new CLOs" — refactored to subobjective level in v2.1 (they are sub-CLO-grain). Reframed decision-grade evidence; renamed domains.

v1 — March 2026 (initial; preserved in git history)

- Original five competencies: Systems Analysis · Iterative Development · Test and Evaluation · Navigating Sociotechnical Constraints · Machine Teaming and Adaptation, with CLOs CLO-1 through CLO-5 in that order.
- Retrievable via `git log --follow lens_program/2_LENS_Objectives_Course_Mapping.md`.

DOC 3 · EDITOR BIOGRAPHIES

LENS • Learning Engineering for Next-Generation Systems
About the Editors
Editor biographies — standalone, for iteration. Revised June 2026.
The casebook's central argument, that capability engineering is the applied form of the educational sciences, taught from the School of Education and informed by operational practice, is embodied in its two editors. Each has spent a career working in one half of that pairing and a substantial portion of it in the other. Each contributes to the five pillars on which LENS rests: Mission Literacy, JHU Ecosystem, Intersectional Expertise, Capability Focus, and Flywheel Iteration.
William Gray-Roncal, Ph.D.
Applied Research · Learning Engineering · STEM Workforce
William Gray-Roncal, Ph.D., is Principal Research Engineer at the Johns Hopkins University Applied Physics Laboratory, where he applies systems engineering methods to challenges spanning defense, space, biomedical, and learning systems. His research, sponsored by DARPA, IARPA, NIH, and DoD, bridges computational neuroscience, artificial intelligence, and large-scale data infrastructure. He developed the College Prep and CIRCUIT workforce-development models, through which he has mentored nearly 500 students, and his current work develops quantitative measures of human cognitive performance in operational contexts. He leads the development of the LENS (Learning Engineering for Next-Generation Systems) specialization within the Johns Hopkins Master of Education in Learning Design and Technology, the program directed by James Diamond at the School of Education. Gray-Roncal holds a Ph.D. in Computer Science from Johns Hopkins, an M.S. in Electrical Engineering from the University of Southern California, and a B.S. in Electrical Engineering from Vanderbilt.
On LENS. His CIRCUIT and MERIT workforce models are learning engineering in practice: instrumented, evidence-driven capability development run as iterative cycles against real research missions. In the casebook's terms he carries Mission Literacy and the Practice Flywheel, and he holds the operational and systems-engineering half of its intersectional expertise.
James Diamond, Ph.D.
Learning Sciences · Educational Design · Evaluation
James Diamond, Ph.D., is Assistant Professor in the Johns Hopkins University School of Education and Program Director of the Master of Education in Learning Design and Technology, where he applies design-based research and learning-sciences methods to the design and evaluation of technology-mediated learning environments. His research, supported by the National Science Foundation, the Institute of Education Sciences, the Bill & Melinda Gates Foundation, the MacArthur Foundation's HASTAC Digital Media and Learning Initiative, the Robin Hood Learning + Technology Fund, the Wellcome Trust, and the UK Economic and Social Research Council, bridges digital games and simulations for learning, digital badges and micro-credentialing, classroom technology integration, and disciplinary literacy in history, social studies, civics, and STEM. Before joining Johns Hopkins, he was Senior Research Associate at the Education Development Center's Center for Children and Technology in New York City, where he led projects including Zoom In and the Who Built America? Teacher Mastery Badge System, contributed to Mission US and Possible Worlds, and taught at New York University as an adjunct. As Program Director, he oversees the LDT program within which the LENS specialization is offered. Diamond holds a Ph.D. in Educational Communication and Technology from New York University, an Ed.M. in Educational Technology from Boston University, and a B.A. in History from Boston University.
On LENS. His design-based research and evaluation tradition supplies the method the casebook runs on: cases are diagnosed, paired with a learning-engineering response, and tested in studio against students' own domains. He anchors the JHU School of Education ecosystem and the learning-sciences and measurement half of the program's intersectional expertise. Between the two editors the five pillars are accounted for, and the fifth, Capability Focus, is the one they hold in common: the conviction that what people can do inside a system is a thing you can design, measure, and build, and the reason the program exists at all.
These biographies reference careers documented in publications, program reports, faculty pages, and the curricular materials of the LDT program and the LENS specialization.

Changes from the casebook version, and one item to verify

Removable LENS ties. Each bio ends with an “On LENS” block, set off by a teal left rule, mapping the editor to the casebook's five pillars and to its thesis that capability engineering is the applied form of the educational sciences. These are designed to lift out cleanly: delete the two “On LENS” paragraphs and the career narratives stand on their own, unchanged. Keep them where the bios accompany the casebook or LENS materials; drop them for a generic faculty or conference bio.

Role framing. “Leads the LENS specialization” is now “leads the development of the LENS specialization within … the program directed by James Diamond.” This names Jim's authority and SOE's ownership while crediting Will's development role; it removes the standing-authority claim the earlier phrasing carried. Diamond's bio adds one sentence making the oversight relationship explicit, so the two read as a matched pair.

Research phrasing. “Currently pursuing research on quantitative measures of human cognitive performance in operational contexts” is now “his current work develops quantitative measures …” If any part of this involves human subjects, confirm IRB determination before the line stays in a published volume; soften to methods/instrumentation language if not yet cleared.

Verify before print. The “nearly 500 students” figure now appears in print; confirm it is current and citable.

Minor copyedits: spelled out USC, normalized em-dashes to commas per house style.

DOC 4 · LENS / LECF CROSSWALK

LENS and the Learning Engineering Competency Framework
Crosswalk, Evidence Status, and Recommended Revisions to LENS Competency Domains
Working document for the ICICLE Competencies SIG conversation. Internal draft; Chatham House attribution conventions apply.

DOCUMENT SECTION

Purpose

This document maps the five LENS competency domains against the proposed Learning Engineering Competency Framework (LECF), the nine-domain synthesis prepared for the IEEE standards pathway. It does three things: states the structural relationship between the two frameworks; provides a domain-level crosswalk with evidence status; and recommends specific revisions to LENS objectives. LENS objectives are revisable. The LECF targets field consensus and moves on a slower, evidence-gated track. Where the two pull against each other, LENS should adjust.

DOCUMENT SECTION

The Structural Claim

LENS is a domain specialization and proficiency claim within the LECF, not a competing framework. Using the INCOSE analogy: the LECF defines what the field's competencies are; LENS trains practitioners to practitioner or lead-practitioner level in a subset of them, within high-consequence operational contexts, and adds domain-specific competencies the way INCOSE accommodates domain extensions. This single sentence carries both audiences. ICICLE sees LENS as an implementation site for a shared framework; SOE sees a program grounded in an emerging IEEE standard.

DOCUMENT SECTION

Crosswalk

Evidence tags follow the LECF convention: [V] validated; [P] practice-grounded, no formal validation; [A] aspirational/proposed, requires new validation work.

LENS Domain
LECF Mapping
Relationship
Evidence
Notes
LENS 1: Systems Thinking & Analysis
LECF 3 (Engineering & Systems Thinking), esp. 3.4 requirements & V&V
Extends
[P] / [A]
Adds capability requirements translation across the human-machine interface; no LECF counterpart
LENS 2: Learning Engineering Design
LECF 1 (Learning Sciences), 2 (Human-Centered Design), 6 (Process & Iteration)
Aligns
[P]
Shared field consensus territory; cleanest mapping
LENS 3: Data, Measurement & Evaluation
LECF 4 (Instrumentation), 5 (Analytics), 7 (Performance & Outcome Engineering)

Aligns + carries proposed extension
[V] / [P] / [A]
Strongest validation in both frameworks; gap attribution and outcome coupling are LECF Domain 7, flagged
[A]
LENS 4: Context & Domain Fluency
No LECF analog
Diverges (new)
[A]
LECF is deliberately domain-agnostic; LENS makes domain fluency a competency; needs behavioral definition (see R1)
LENS 5: Emerging Systems & Futures
LECF 5.2 (responsible AI/ML), partial
Extends
[A]
Human-AI teaming and not-yet-fielded systems are absent from LECF and ICICLE/LEBOK
LENS ethics thread (distributed)
LECF 8 (Professional, Ethical & Collaboration Practice)
Aligns; different architecture
[P]
LECF houses ethics as a domain; LENS distributes it as a design constraint across all courses

DOCUMENT SECTION

What the Crosswalk Shows

LENS takes the LECF's least-validated territory and makes it the center of gravity. Domains 2 and 3 sit on consensus ground with the field's strongest evidence. Domains 1, 4, and 5, plus the gap-attribution and outcome-coupling material in Domain 3, are extensions: intellectually defensible, grounded in HSI and human performance technology, and unvalidated in learning engineering specifically. This is not a weakness to hide; it is the program's argument. But it sets two obligations. First, external documents must label extensions as proposed, never as field consensus. Second, LENS itself must generate the validation evidence, which is the subject of R3 below.

DOCUMENT SECTION

Recommended Revisions to LENS Objectives

These take advantage of the stated flexibility in LENS objectives. Each is concrete enough to act on before the SIG conversation.

R1. Rewrite Domain 4 with behavioral definitions

Current Domain 4 language ("high-consequence operational contexts, HSI frameworks, interdisciplinary team engagement") names content, not competence. It is the only LENS domain that cannot survive the question: what does a person observably do that demonstrates this? Proposed replacement subcompetencies:

- **4.1** Translate domain constraints (regulatory, safety, operational tempo, security classification) into learning design requirements
- **4.2** Conduct stakeholder and context analysis in a high-consequence environment using an HSI framework, producing a documented context model
- **4.3** Communicate learning engineering decisions in the working vocabulary of a host domain, verified by domain stakeholder review

Each of these is assessable from capstone artifacts. The current language is not.

R2. Declare the individual-attainment posture explicitly

The LECF follows the INCOSE convention: competencies a learning engineering capability must cover, often distributed across a team. LENS makes individual claims: competencies a graduate demonstrates. These are compatible but the crossing must be explicit. Recommended move: publish a one-page attainment profile stating which LECF subcompetencies a LENS graduate reaches at practitioner level, which at awareness level, and which are out of scope. The IBSTPI precedent (individual competencies validated against practitioners) supports this posture; adopt it by name in the SIG conversation.

R3. Build the Domain 7 validation pathway into LENS, with external sites

LECF Domain 7 (gap attribution, learning-to-operational-outcome coupling, transfer specification) is tagged [A] throughout and is the material most identified with this effort. LENS capstones are the natural evidence engine: rubrics scoring gap-attribution quality, documented coupling analyses, transfer measurement plans. One constraint is non-negotiable: if the only evidence for Domain 7 comes from LENS, it is self-validation. Recruit at least one or two external pilot sites through the SIG and the broader ICICLE community to generate independent practice evidence. This is also the strongest collaborative offer available for the SIG conversation: LENS contributes its evidence and invites the community to contribute theirs.

R4. Anchor Domain 5 to the LECF and name the candidate extensions

LENS Domain 5 currently floats. Anchor it to LECF 5.2 (responsible AI/ML) and propose two named candidate subcompetencies for SIG debate:

Design capability development for human-AI teams, where task allocation between human and machine is itself a design variable

Specify learning and capability requirements for not-yet-fielded systems, working from system design artifacts rather than operational experience

Framing them as proposals for the field, rather than LENS-only content, converts a divergence into a contribution.

R5. Keep ethics distributed; add the mapping statement

The architectural difference (LECF houses ethics as Domain 8; LENS threads it through every course) is a strength, consistent with the principle that deferred commitments do not materialize. The only fix needed is documentation: a short table stating where each LECF 8.x subcompetency is taught and assessed across LEN 1 through the capstone. Without it, reviewers will read the distributed approach as absence.

DOCUMENT SECTION

Domain Language: Current vs. Recommended

Domain	Current language	Recommended revision
LENS 1	SE principles, human-system task analysis, capability requirements translation	Keep. Add explicit mapping language: “extends LECF Domain 3 with requirements translation across the human-machine interface.” Propose 3.5 as a candidate LECF subcompetency.
LENS 2	Learning sciences application, evidence-based design at scale, transfer to high-stakes environments	Keep; split the transfer clause. “Transfer to high-stakes environments” belongs with LECF 7.3 (transfer specification), not with design. Moving it consolidates all [A] material in one place.
LENS 3	Ethical instrumentation, learning analytics, operational performance criteria, gap attribution	Keep; label gap attribution and operational performance criteria as “proposed extensions to the field framework (LECF Domain 7)” in any external document.
LENS 4		

High-consequence operational contexts, HSI frameworks, interdisciplinary team engagement
Rewrite with behavioral definitions (R1). Current language is a knowledge area dressed as a competency.
LENS 5
Human-AI teaming, capability development for not-yet-fielded systems, adaptive learning at scale
Keep; anchor to LECF 5.2 and propose two named candidate subcompetencies for SIG debate rather than leaving them free-floating (R4).

DOCUMENT SECTION

Framing for the SIG Conversation

Lead with credit: the Learning Engineering Process model, the Toolkit, and LEBOK define the knowledge map. The LECF proposes the behavioral competency layer LEBOK invites; LENS is a first implementation site.
Offer evidence, not territory: LENS capstone artifacts as documented practice for Domain 7, contingent on external pilot sites to avoid self-validation.
Hold the proficiency-level proposal in reserve: INCOSE-style levels per subcompetency are the highest-leverage next step for the framework, and the LENS attainment profile (R2) is a ready-made demonstration.
Do not attach credential or certification language to anything yet. The evidentiary bar (job-task analysis under ISO/IEC 17024 logic) is not met, and claiming otherwise costs credibility with exactly the audience that matters.
In plain language
The field framework says what every learning engineer should be able to do. LENS says: here is a program that builds those abilities to a professional level, in places where mistakes are expensive, and adds a few abilities the field has not yet agreed on. We are honest about which is which, and we are offering our students' work as part of the evidence the field needs.

DOC 5 · SUMMARY AND PRE/POST OBJECTIVES

From the 2019 Competencies Paper to a Standards-Ready Framework
One-page discussion summary
The 2019 paper set the agenda and deferred the work. The LECF and LENS execute its stated next step.
The foundation. Goodell, Kessler, Kurzweil and Kolodner (2019), "Competencies of Learning Engineering Teams and Team Members," organizes competence as five shared dispositions (focus on learning and learners; reflective practitioner and iterative refiner; responsive to data, evidence, research, and requirements; leverage systems thinking; mindful of context), three process groundings, and a flower of contributing disciplines. It contains no competency table and no role matrix; it closes by calling for exactly this: "operationalize these organizing principles into sets of specific competency definitions."
The lineage claim. The LECF re-expresses the LEBOK knowledge map as assessable, behaviorally defined competencies. LENS implements them as a graduate concentration in high-consequence contexts. Neither competes with the 2019 paper or with LEBOK; both complete the paper's stated next step.
Domain
Relationship to 2019
Basis
LECF 1, 2, 6 (learning sciences, design, process)
Direct antecedent
The paper's three groundings and the iterative-refiner disposition
LECF 3, 4, 5 (systems, instrumentation, analytics)
Partial antecedent
Systems thinking as disposition; data-informed decisions as grounding; depth came later via Toolkit and LEBOK
LECF 8, 9 (professional practice, operations)
Partial / weak
Collaboration thesis is core to the paper; ethics and project management entered the canon later
LECF 7 (performance and outcome engineering)
New extension
No gap attribution, no Gilbert, no outcome coupling in 2019; flagged [A], needs validation
LENS 4 (context and domain fluency)
New extension
"Mindful of Context" disposition is the nearest 2019 element; HSI and high-consequence framing is new
LENS 5 (human-AI teaming, future systems)
New extension
Pre-generative-AI paper could not address it; vintage is the rationale, not a critique
The one tension, resolved. The 2019 thesis is that the team is the unit of competence; no individual holds it all. LENS makes individual claims. Resolution: LENS certifies individual cross-domain fluency sufficient to lead and integrate a learning engineering team. The team remains the unit of delivered capability; the credential names what one integrator contributes to it.
What we propose to carry forward. Re-surface the five dispositions as cross-cutting professional practice in any standard; preserve the flower's extensibility ("and other fields as the challenge requires"); label every extension as proposed, never as consensus; sequence validation honestly (Delphi and card-sort for structure; job-task analysis before any credential claim).
The offer. LENS contributes capstone artifacts as documented practice evidence for the unvalidated extensions, contingent on external pilot sites so the evidence is not self-validating. The framework is a map, not a gate: adoption through usefulness, with credentialing deferred until the evidentiary bar is met.
LENS Concentration Objectives: Entering and Exiting (Proposed)
Stated as observable pre/post pairs to support measurement of the concentration's effect
Posture. Entering students arrive grounded in one or two disciplines of the field's flower; that is expected, consistent with dual accessibility for education professionals and engineers alike. The concentration's claim is the delta: graduates demonstrate practitioner-level competence in the core domains and integration fluency across all of them. Each pair below is written so the pre state is genuinely common at entry and the post state is assessable from course and capstone artifacts.
Entering students can (pre):

Graduates can (post):
Describe how people learn in at least one professional tradition (teaching, training, instructional design, or an engineering discipline)
Apply learning sciences principles to design decisions and state each design choice as a testable hypothesis
Identify stakeholders in a familiar setting and describe their needs informally
Conduct stakeholder and context analysis in a high-consequence environment using an HSI framework, producing a documented context model
Recognize that learning solutions exist inside larger organizational and technical systems
Model a learning environment and its host operational system, translating capability requirements across the human-machine interface into learning design requirements
Interpret simple descriptive data about learners (completion, scores) presented by others
Instrument learning environments ethically and produce decision-grade evidence: dashboards, reproducible reports, and analyses that link learning measures to operational performance criteria
Attribute performance problems to training by default
Diagnose whether a capability gap is rooted in learning, system design, or organizational conditions before proposing an intervention (gap attribution; proposed field extension)
Use AI tools as a consumer
Design capability development for human-AI teams and specify learning requirements for not-yet-fielded systems from design artifacts
Work within their own professional silo, handing off at its boundary
Lead and integrate a learning engineering team: collaborate across disciplines with shared vocabulary, exhibiting the field's professional dispositions

Notes for review

Pairs 1 through 4 and 7 sit on practice-grounded field territory (LECF Domains 1 through 6, 8). Pairs 5 and 6 are proposed extensions (LECF Domain 7; LENS Domains 4 and 5) and are labeled as such in any external document.
Pair 5's pre state is deliberately blunt: defaulting to a training fix is the documented norm the concentration exists to break. If that reads as unfair to incoming students, soften the pre side, not the post side.
Ethics is not a pair because it is not a delta: ethical instrumentation and equity considerations are design constraints embedded in pairs 2, 4, and 7 from the first course onward.
The pairs double as a pre/post assessment plan: each pre state can be measured at entry (intake survey, diagnostic scenario) and each post state from rubric-scored artifacts, giving the concentration its own decision-grade evidence of effect.
In plain language
Students arrive knowing one part of the picture. They leave able to see the whole system, prove what works, tell a learning problem from a system problem, and lead the team that fixes it.

DOC 6 · RECRUITMENT EMAIL

LENS • Learning Engineering for Next-Generation Systems
Student Recruitment Email
Two variants for different audiences. Revised June 2026.
Send the mission-led variant to individuals who fit the LENS profile directly. Send the credential-led variant to network contacts, referrers, and list outreach where forwarding is likely. Attach the program overview and competency framework; for warm prospects, include a casebook excerpt.
Variant A · Mission-led (direct to a prospective student)
Subject: A new way to build what people can do inside complex systems
Dear [Name],
Systems don't fail because the technology breaks. They fail because someone couldn't do what the system required, and almost no one engineers for that.
We are launching LENS, Learning Engineering for Next-Generation Systems, a new specialization within the Johns Hopkins School of Education's Master of Education in Learning Design and Technology. It starts in August 2026, online and part-time, built for working professionals.
LENS treats human capability as a systems engineering problem. Across healthcare, defense, and education, the question is the same: what must people be able to do for the system to deliver its impact, and how do you design, measure, and sustain that at speed and scale? Our graduates learn to answer it with decision-grade evidence, not opinions about training.
The specialization is built around five competencies: seeing the whole system, building and refining interventions iteratively, showing what works through measurement, making solutions survive real-world constraints, and teaming effectively with AI. You can find the full competency framework and curriculum in the materials attached.
Who it is for: you do not need to arrive knowing all of this. If you come from teaching, training, engineering, healthcare, or operations and you want to learn to build capability as an engineered, evidence-grounded discipline, LENS is designed for you. No one carries every tool; this is team work, and what you already bring is the place you start.
If you would like to learn more, I would welcome a conversation. The attached casebook excerpt and program overview will give you a feel for how we think.
With best regards,
William Gray-Roncal, PhD
Learning Engineering for Next-Generation Systems
Johns Hopkins University School of Education
[email] | capabilitymatters.org
Variant B · Credential-led (network, referrers, list outreach)
Subject: New at Johns Hopkins: a Learning Engineering specialization (Fall 2026)
Dear [Name],
I am writing to share a new graduate specialization that may interest you or people in your network.
This fall, the Johns Hopkins School of Education launches LENS, Learning Engineering for Next-Generation Systems, a specialization within the online Master of Education in Learning Design and Technology. It is part-time and built for working professionals, and it is the first program to bring learning engineering, human systems integration, and high-consequence operational domains together as one discipline.
The premise is simple to state. Every complex system depends on what people can do inside it, that dependency is measurable and designable, and it is too important to leave to chance. LENS prepares practitioners to engineer it: to define capability requirements, build evidence-based interventions, measure their operational impact, and sustain them across the system lifecycle, in healthcare, defense, and education alike.
Graduates demonstrate five competencies: systems analysis, iterative development, test and evaluation, navigating sociotechnical constraints, and machine teaming and adaptation. The program is ten courses and thirty credits, with a capstone run on a real operational problem. Full details, including the competency framework and course map, are attached.

The specialization welcomes two audiences equally: education and training professionals moving toward systems practice, and engineers and technical professionals moving toward learning design. Founding-cohort enrollment is open now for August 2026.

I would be glad to talk through fit, curriculum, or the application process. Please feel free to forward this to anyone it might serve.

With best regards,

William Gray-Roncal, PhD
Learning Engineering for Next-Generation Systems
Johns Hopkins University School of Education
[email] | capabilitymatters.org

Verify before sending: confirm the August 2026 start, online/part-time format, and “founding cohort” language against current admissions copy; confirm the email address and that capabilitymatters.org is live. Per program convention, defense is named as one domain among three, never the primary one.

COMPANION TO

Capability Matters: A Casebook

Edited by William Gray-Roncal, PhD and James Diamond, PhD

Learning Engineering for Next-Generation Systems · v2.1 · Edition · 15 June 2026

LDT · School of Education · Johns Hopkins University